

Loan Approval Prediction Executive Summary

This one-page executive summary captures the key findings of the loan approval prediction project, including the workflow, model selection, and visual evidence from the notebook analysis.

- The notebook follows a complete machine learning pipeline: data loading, EDA, missing value treatment, feature encoding, scaling, SMOTE-based class balancing, and model evaluation.
- The dataset shows a moderately imbalanced approval distribution, so evaluation focused on precision, recall, F1-score, and ROC-AUC rather than accuracy alone.
- Credit history emerged as one of the strongest predictors of approval, while income and loan amount also influenced the outcome.
- Logistic Regression was selected as the final model because it provided the best overall balance of performance and interpretability for this business use case.

Model description

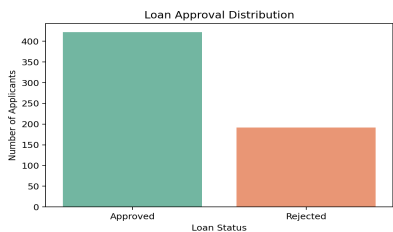
A Logistic Regression classifier was trained after preprocessing categorical variables with one-hot encoding and scaling numerical features. The training data was balanced with SMOTE to improve classification performance on the minority class.

Notebook and dataset references

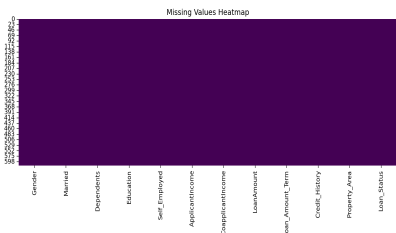
Notebook: Loan/Loan_Approval_Prediction.ipynb

Dataset: Loan/loan_prediction.csv

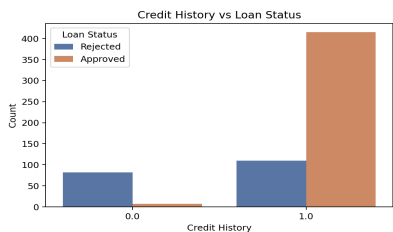
Loan Approval Distribution



Missing Values Heatmap



Credit History vs Loan Status



ROC Curve - Logistic Regression

